ROSHAN MOHAMMED

AI / Machine Learning Engineer | RAG · LangChain · FastAPI · AWS

Full Work Rights in Australia | No Employer Sponsorship Required

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Live Demo: documind.mroshan454.dev | LinkedIn | GitHub | Hugging Face

SUMMARY

Melbourne-based AI/Software Engineer with an MSc in Artificial Intelligence, specializing in building and deploying production-ready LLM and RAG applications. Recently shipped DocuMind, a live RAG system on AWS (FastAPI, Docker, Nginx, HTTPS). Hands-on across the modern AI stack — LangChain, vector databases, OpenAI integration, prompt engineering, and model evaluation — with strong production-engineering discipline, including an 89% reduction in production Docker image size. Combines applied AI delivery with deep ML foundations.

TECHNICAL SKILLS

AI Engineering: LangChain, ChromaDB (vector store), OpenAI API, RAG architecture, embeddings, prompt engineering, RAG evaluation (faithfulness, recall@k)

Languages: Python (advanced), SQL, Bash, JavaScript (basic)

Backend / APIs: FastAPI, REST APIs, Pydantic, async/await, Uvicorn, Swagger/OpenAPI

DevOps / Cloud: Docker, AWS EC2, Nginx (reverse proxy), Let's Encrypt SSL, Cloudflare DNS, Linux, SSH, Git, GitHub Actions

ML / Deep Learning: PyTorch, Transformers, CNNs, Vision Transformers, transfer learning

Testing / Observability: pytest, structured logging (structlog), custom exception handling, mocking

PROJECTS

DocuMind — Production RAG System on AWS*Mar 2026 – May 2026*

FastAPI · LangChain · ChromaDB · OpenAI · Docker · AWS EC2 · Nginx · Let's Encrypt · Cloudflare

Live: documind.mroshan454.dev/docs | *GitHub:* github.com/mroshan454/documind

- Built and deployed an end-to-end document-QA backend: PDF ingestion → vector indexing → LLM-grounded answers with cited sources. Publicly accessible via HTTPS on a custom domain.
- Deployed a containerized FastAPI service on AWS EC2 with Nginx reverse proxy, Let's Encrypt SSL (auto-renewing), Cloudflare DNS, and a static Elastic IP.
- Reduced the production Docker image from 8.9 GB to under 1 GB (≈89%) by migrating from local HuggingFace embeddings to OpenAI's embedding API after profiling real dependencies.
- Designed idempotent ingestion with SHA-256 deterministic chunk IDs; implemented async LLM calls via LangChain ainvoke() for concurrent request handling.
- Built a custom exception hierarchy with structured error codes, structured JSON logging (structlog), and a pytest suite (unit + integration with mocked LLM/vectorstore).

CrossSight — Multimodal Image Captioning System*Jan 2026 – Mar 2026*

PyTorch · CNN encoder + Transformer decoder · Cross-Attention · Gradio

Live: huggingface.co/spaces/roshan454/CrossSight | *GitHub:* github.com/mroshan454/CrossSight_Image_Captioning

- Designed and deployed an image-captioning system pairing a CNN encoder with a Transformer decoder via cross-attention; live demo on Hugging Face Spaces.

PlantNet39 — Plant Disease Classifier (39 classes)*Sep 2025 – Dec 2025*

EfficientNet-B2 · Transfer Learning · PyTorch · Gradio

Live: huggingface.co/spaces/roshan454/PlantNet39 | *GitHub:* github.com/mroshan454/PlantNet39

- Built and deployed a 39-class plant-disease classifier via transfer learning on EfficientNet-B2; ~35 MB model optimised for CPU-only inference.
- Added a dedicated non-leaf background class to reject invalid inputs gracefully; shipped a live public demo on Hugging Face Spaces.

Deep Learning Foundations — GPT & ViT from Scratch*2025 – 2026*

PyTorch · No high-level transformer libraries

GPT: github.com/mroshan454/GPT-From-Scratch-PyTorch | *ViT:* github.com/mroshan454/Replicating-ViT-Research-Paper

- Implemented a GPT-style transformer and replicated the Vision Transformer paper end-to-end in PyTorch — tokenization, embeddings, multi-head self-attention, causal masking, patch/positional embeddings — demonstrating deep architectural understanding beyond library usage.

EXPERIENCE

Research Assistant (Contract) — Energy Consumption Prediction *May 2024 – Jul 2024*

Buckinghamshire New University, United Kingdom

- Developed energy-consumption prediction models on UK EPC datasets (scikit-learn); applied SHAP for interpretability; delivered an internal web demo.

Teaching Assistant — Python Programming *Oct 2024 – Mar 2025*

Buckinghamshire New University, United Kingdom

- Supported undergraduate Python workshops; mentored first-year students on Git, debugging, and assignment reviews — strong technical communication to non-experts.

EDUCATION

MSc Artificial Intelligence*2023 – 2024*

Buckinghamshire New University, United Kingdom

BSc Computer Science*2020 – 2023*

Kerala University, India